Algorithm for file updates in Python

Project description

In this project, I am serving as a security professional working at a healthcare company. My job is to regularly update a file that identifies the employees who can access restricted content. The `"allow_list.txt"` file showcases these employees based on their IP addresses. There is also a remove list that identifies the IP addresses of employees who should no longer be able to access the content. I have created an algorithm that reads an existing allow list file, compares it to a remove list containing IP addresses that should no longer have access, and removes any matches.

Open the file that contains the allow list

First, I began to open the `"allow_list.txt"` file by assigning `import_file` to a string containing the file name.

```
# Assign a string containing the file name to 'import_file'
import_file = "allow_list.txt"
```

Then, I utilized a `with` statement to open the file. The statement below showcases which file I want to open (indicated by `open(import_file,`) and that I want to read the file (showcased by `"r"`). I also utilized the `as file:` to assign the open file object to the variable `file`.

```
# Utilize 'with' to enable reading of file contents
with open(import_file, "r") as file:
```

Read the file contents

Next, I utilized `.read()` to allow for reading of the entire contents of the file as a single string stored in `ip_addresses`.

```
# Utilize 'with' to enable reading of file contents
with open(import_file, "r") as file:
    ip_addresses = file.read()
```

Convert the string into a list

I then utilized the `.split()` method to convert the ip_addressess string into a list and reassigned it to the variable `ip_addresses` to store it.

```
# Utilize 'split()' to convert from string to list
ip_adresses = ip_addresses.split()
```

Iterate through the remove list

Furthermore, I created a for loop to iterate through the elements in `remove_list`. This loop will repeat a code to do a specific action I specify. The statement below will iterate through `ip_addresses` and assign each value to the loop variable `element`.

```
# Create for loop to iterate through the remove list
for element in remove_list:
```

Remove IP addresses that are on the remove list

Next, I created a conditional statement that evaluated if `element` was found in `ip_adresses` list. If I did not specify this, then trying to remove elements not found in the `ip_addresses` list would showcase an error.

Then I wrote a statement using `.remove()` to indicate that if `element` was found, remove it from `ip_addresses` list.

```
# Create for loop to iterate through the remove list
for element in remove_list:
    # create conditional statement to evaluate whether 'element' is in 'ip_addresses'
    if element in ip_addresses:
        # use '.remove()' to remove 'element' from 'ip_addresses'
        ip_addresses.remove(element)
```

Update the file with the revised list of IP addresses

Additionally, I created a statement to convert `ip_addresses` back to a string in order for it to be written into the text file using `.join()`. The `\n` used in the statement indicates to the program to place each element on a new line.

```
# Covert back into string
ip_addresses = "\n".join(ip_addresses)
```

Finally, I used another `with` statement to open the file and the `.write()` to update it.

```
# Rewrite file using 'with' statement
with open(import_file, "w") as file:
    # Replace file contents with 'ip_addresses'
    file.write(ip_addresses)
```

Summary

The algorithm I created works to update the `allow_list.txt` file by removing IP addresses that are no longer authorized to access the content. It opens and reads the allow list file, converting its contents from a string into a list. Then it iterates through a for loop that checks each item in the remove list and removes any matching IP addresses from the allow list. Once it removes the IP address, it converts back to a formatted string and overwrites the original file. This successfully showcases my knowledge with file handling, iteration, and conditional logic in Python.